

---

Founding Engineer, Brown Alum with a specialization in Computer Graphics. I'm at my best tackling complex technical puzzles and leading ambitious projects, and love collaborating with teams that share my passion!

---

## LANGUAGES & FRAMEWORKS

- Specialties in C++, Vulkan, OpenGL, Python, ThreeJS, React, Typescript, experience with C, Rust, Go, Java, GCP
- 

## EXPERIENCE

**Saga Graphics Lead, Founding Engineer**, Feb 2025-Current (AI-powered Professional Production/Events Platform)

- Owned architecture for platform's core realtime 3d features, writing production code for 5k+ active users
  - Engineered custom state synchronization of a complex WebGL context with the Virtual DOM, enabling bidirectional integration of 3d interactive features with UI, and maintaining optimal 60fps even on low compute resources
  - Implemented explicit memory lifecycling, custom MVP matrix controllers, and efficient AABB raycasting
  - Primary tech/product liaison, selected for my ability to empathize with and effectively communicate each team's needs
- 

## PROJECTS

**VulkaN64: A Modern Vulkan 1.4 Rendering Engine**, 2026

- Built Vulkan engine from scratch w/ GPU profiling, VMA, Compute Shaders, Sync2, and Dynamic Rendering
- Achieved 0.05ms frame times (20k FPS) w/ <0.01ms sync overhead, incl. 3D particle and post-process compute passes
- Utilized comprehensive graphics/systems background for rapid 1.5 month onboarding/development time
- Skills: C++, Vulkan, Slang, CMake, Particle Systems, GPU Programming

**Authoring Landscapes w/ Ecosystem and Terrain Erosion Simulation**, 2024

- Built upon modern Computer Graphics research papers on ecological phenomena, implementing several concurrently
- Developed simulations for rainfall, sediment transport, and erosion on layered land model to generate realistic, natural landscapes and rendered using GLSL (OpenGL) shaders
- Skills: Rust, OpenGL, Stochastic Simulation, Procedural Generation, Research Paper Implementation

**OpenGL Procedural Terrain Renderer**, 2024

- Created a ground-up realtime pipeline showcasing volumetric lighting (god rays), procedural terrain generation with dynamic object placement and textures, and a custom Bezier spline camera controller
- Skills: C++, OpenGL, Custom Shaders, Ray Marching, Procedural Generation

**Ray Tracer and Path Tracer**, 2024

- Wrote ray and path tracers, producing photo-realistic rendered images
- Implemented indirect lighting, fresnel refraction, supersampling anti-aliasing, and soft shadows; supported mirror, glossy, and diffuse surfaces, ultimately allowing physically accurate rendering of any material
- Skills: C++, Ray Tracing, Path Tracing, Monte Carlo Integration, Research Paper Implementation

**NeRF Research at Interactive 3D Vision & Learning Lab**, 2023

- Created NeRF pipeline using data from custom multiview camera capture system and camera extrinsics determined via COLMAP to train custom, world-class models in a bleeding-edge field
  - Significantly accelerated via automation project goal of quickly and consistently contributing quality data and models
  - Skills: Python, PyTorch, Deep Networks, 3D Reconstruction, Computer Vision, Language Models, Gen AI
- 

## EDUCATION & AWARDS

**Brown University, Providence, Rhode Island**, graduated June 2024

- Degree: BS Computer Science with Visual Computing and AI/ML Pathways
- Capstone: Year-long Graduate Graphics seminar on theory and implementation of a wide range of modern techniques
- Courses: Computer Vision\*, Systems, Security\*, Linear Algebra, Deep Learning\*, Real Analysis\*, Japanese Lit.\*
- Awards: UTRA Research Grant at Interactive 3D Learning & Vision Lab; Event of the Year, Brown Puzzlehunt
- Teaching Assistant for Computer Vision (Senior/Graduate seminar)

*\*Senior/Graduate seminar*

---

## LEADERSHIP

**Founder & Lead Writer, Brown Puzzlehunt/Brown Puzzle Club**, 2021-2024

- Founded Brown Puzzlehunt (yearly event w/ 1000+ participants) and Brown Puzzle Club
- Personally responsible for content and editing of all (35+) puzzles, managing 20+ writers for each hunt
- Grew the event/club from the ground up, eventually winning Brown's Event of the Year Award

---

## EDUCATION & AWARDS (CONT.)

**Phillips Exeter Academy, Exeter, New Hampshire**, graduated June 2020

- 1570 SAT, Highest Academic Honors (1st Quintile)
- Highest level CS, 4 years Japanese, Senior sem. at Seikei Tokyo, Head of Peer Tutoring, Varsity Cycling & Track

---

## INTERESTS

- Fluent in Japanese, spent 6 months at Keio University in accelerated language program, 8 years total study
- Ongoing obsessions include bouldering, reading manga, Pokemon draft leagues, hyperpop, and FX's The Bear
- Favorite Games: Noita, Legend of Zelda: Ocarina of Time, No Man's Sky, Overwatch, Tetris, Project Winter
- Multi-ethnic/cultural: Lived 1 year in Tokyo, 8 months in London, traveled to 4 continents and 10+ countries